Is Global Warming Just a Trend?

A time series analysis of Berkeley Earth Temperature Anomalies

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Question: Has global warming just an upward trend, or have the patterns of temperature variation changed too?

1. Did the warming rate increase after 1950 or 1980?
2. Are modern temperature anomalies more persistent than early-record anomalies?
3. Is there evidence of 2-7 year periodicity after detrending?
4. Do seasonal or time-varying trend models improve forecasting?

Why time series?

Monthly temperatures are connected over time. Heat stored in the ocean and atmosphere creates persistence, and climate cycles can create multi-year patterns. Time series methods let us separate long-term warming from short-term dependence and possible cycles.

Data: Berkeley Earth Global Temperature Anomalies

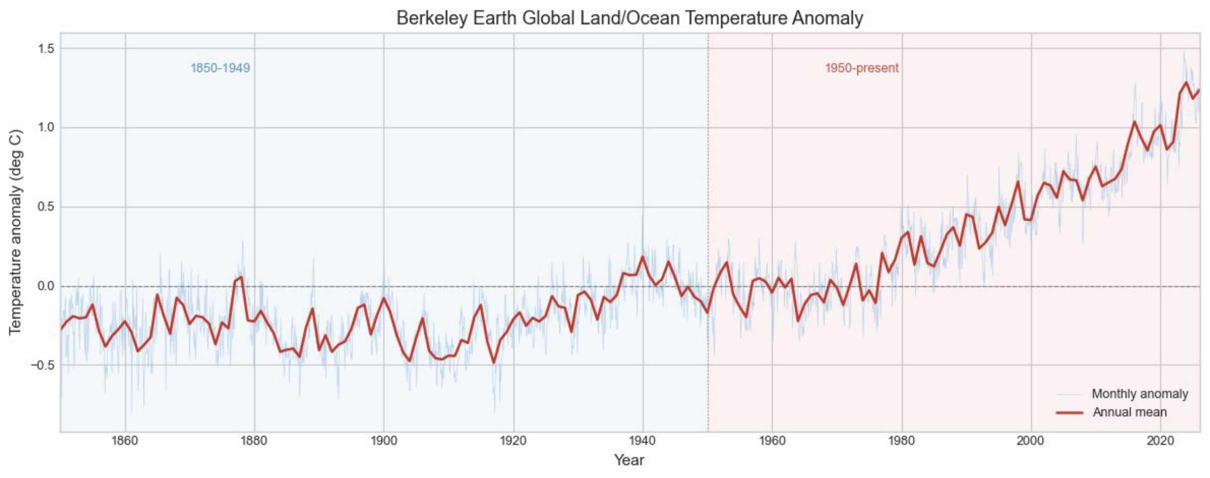
Source: Berkeley Earth Global Average Land/Ocean Temperature Time Series, accessed through NOAA Physical Sciences Laboratory

Coverage

January 1850 to March 2026
2,115 monthly observations
No missing months after cleaning
Min: -0.804℃, Max: +1.477℃

Measurement

Temperature anomaly in degrees Celsius, relative to 1951-1980 baseline



Methods: Trend, Dependence, Periodicity, and Forecasting

1. **Trend modeling**

Compare warming rates across time periods using linear trends

2. **Diagnostics**

Use Augmented Dickey-Fuller (ADF) tests, ACF/PACF plots, first differences, and detrended residuals

3. **Spectral analysis**

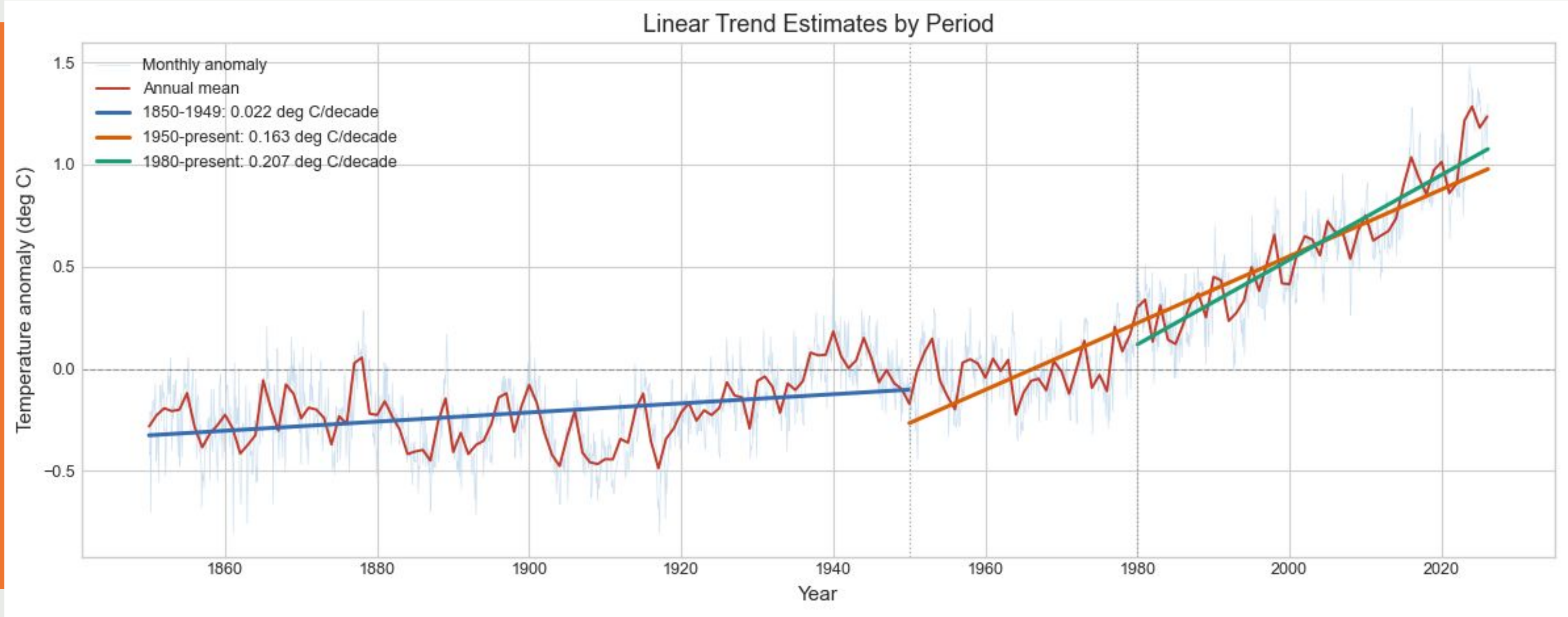
Use Welch's method to examine 2-7 year periodicity after detrending

4. **Forecasting**

Compare ARIMA, SARIMA, and state space models on the 2015-2026 holdout period

Initial trends:

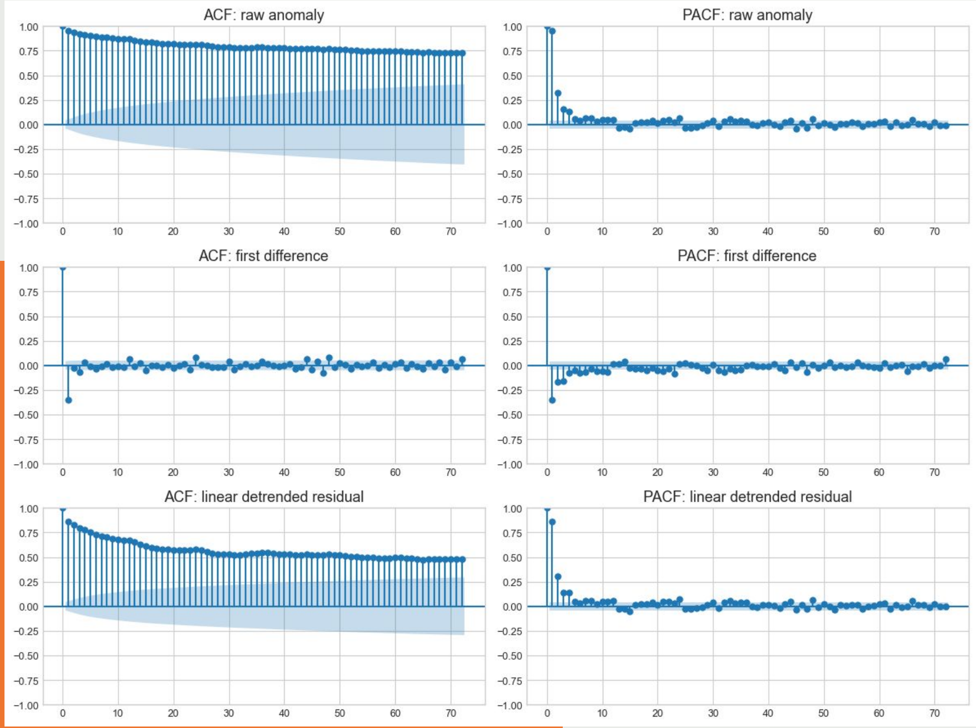
The warming rate increased over time



The post-1980 warming rate is about **9 times** the early-record rate.

Diagnostics:

Trend removal helps, but residual dependence remains

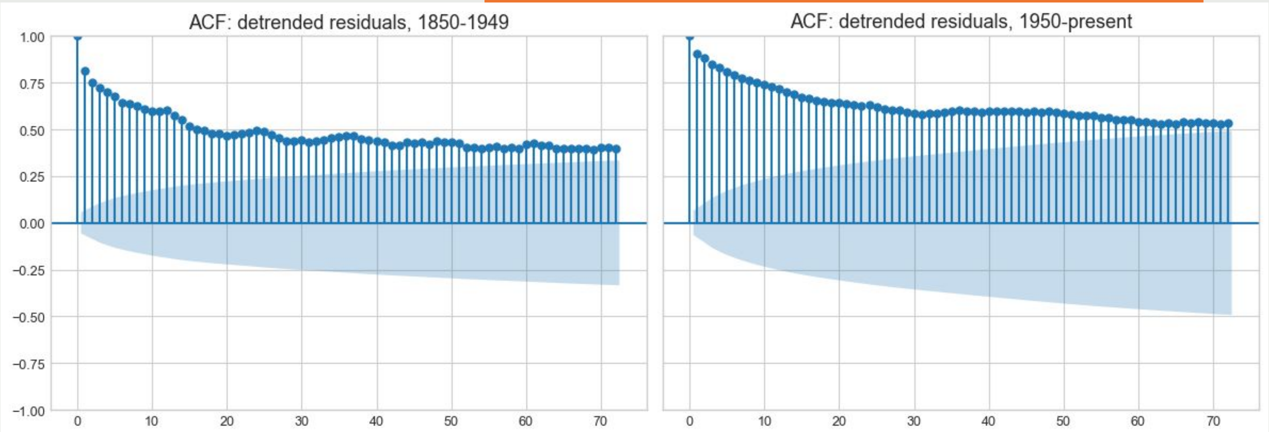


Series	ADF p-value	Interpretation
Raw anomaly	0.944	Not stationary
First difference	< 0.001	Stationary
Linear detrended residual	0.131	Still persistent

Finding 1:

Modern residuals are more persistent

Modern detrended residuals show stronger autocorrelation at short and annual-scale lags.



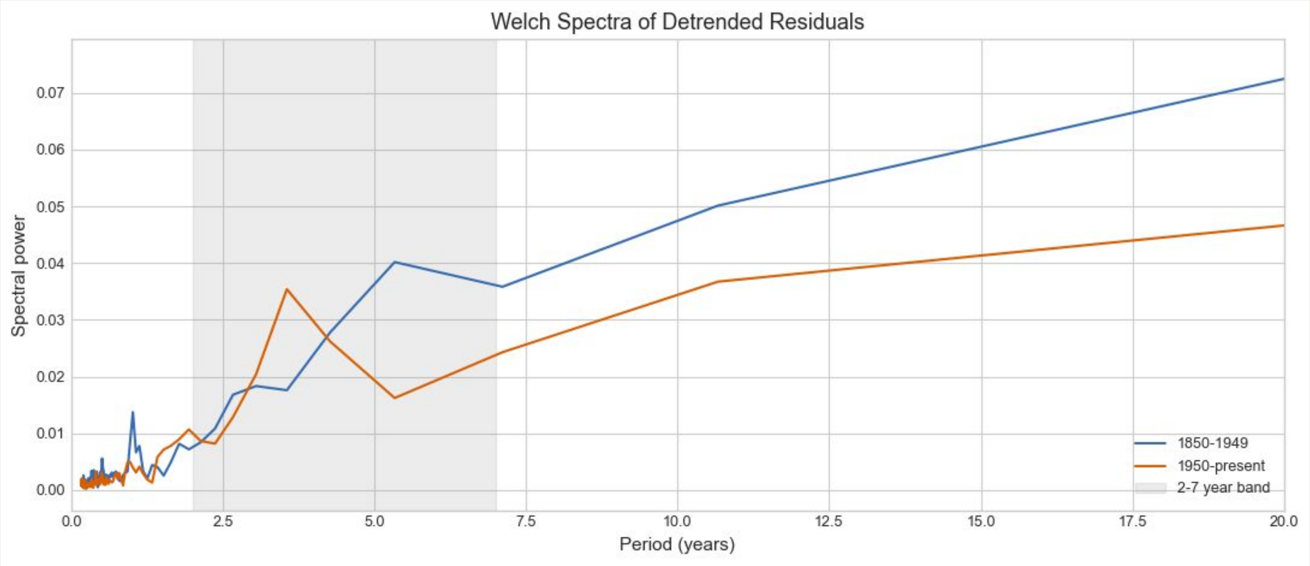
Lag	1850-1949	1950-present
Lag 1	0.814	0.911
Lag 12	0.614	0.744
Lag 24	0.513	0.694

Finding 2:

Spectral analysis shows weak El Niño-Southern Oscillation (ENSO)-Scale Evidence

The modern period does not show stronger average power in the 2-7 year band.

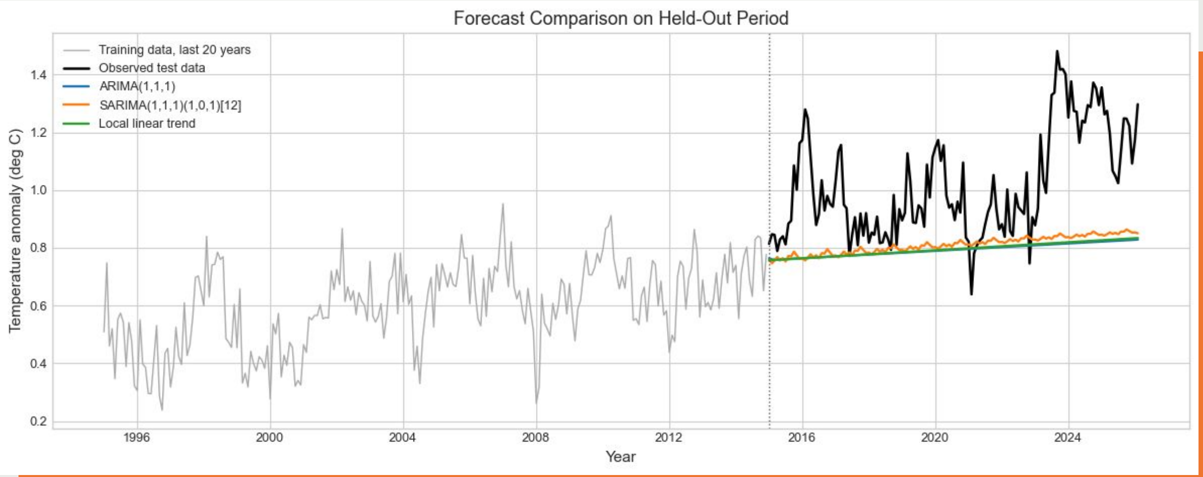
1850-1949: 0.0200
1950-present: 0.0183



Permutation test (n=5,000), two-sided p-value = 0.456

Forecast comparison

SARIMA selected via grid search, slightly outperforms alternatives.



Model	RMSE	MAE
SARIMA(1,1,1)(1,0,1)[12]	0.267	0.213
Local linear trend	0.281	0.226
ARIMA(1,1,1)	0.283	0.228

Model selected by grid search over 35 SARIMA specifications, evaluated on AIC, BIC, and held-out RMSE.

Conclusion

1. **Warming rate increased:** post-1980 slope is much larger than the early-record slope
2. **Modern residuals are more persistent:** modern detrended residuals have stronger autocorrelation
3. **ENSO-scale evidence is weak:** the 2-7 year spectral band was not stronger in the modern period
4. **State space result:** warming is faster after 1980, but the model suggests this looks more like a shift than a smooth gradual change

What this means and what comes next?

Limitations

- Global average may hide regional patterns: ENSO affects regions differently, so the global mean may weaken the signal.
- Trend break was not formally tested: a structural break test could better identify when the warming rate changed.
- Early data may be noisier: measurement uncertainty could make early autocorrelation look weaker.

Bigger picture

- The series is not just warming, modern anomalies also appear more connected over time.
- Better models should separate long-term warming from natural climate variation.
- **Next steps:** Compare regional temperature series, add direct climate variables like CO_2 and an ENSO index, use Berkeley Earth uncertainty estimates

Thank You!

References

- Rohde, R. A. and Hausfather, Z.: The Berkeley Earth Land/Ocean Temperature Record, Earth System Science Data, 2020.
- National Centers for Environmental Information. ENSO: Recent Evolution, Current Status and Predictions. NOAA. https://www.ncei.noaa.gov/access/monitoring/enso/
- Shumway & Stoffer: *Time Series Analysis and Its Applications: With R Examples.* 2017